

EachMovie: archaeology of the/a first latent recommender system

ACM RecSys 1996→2026
Past, Present, and Future

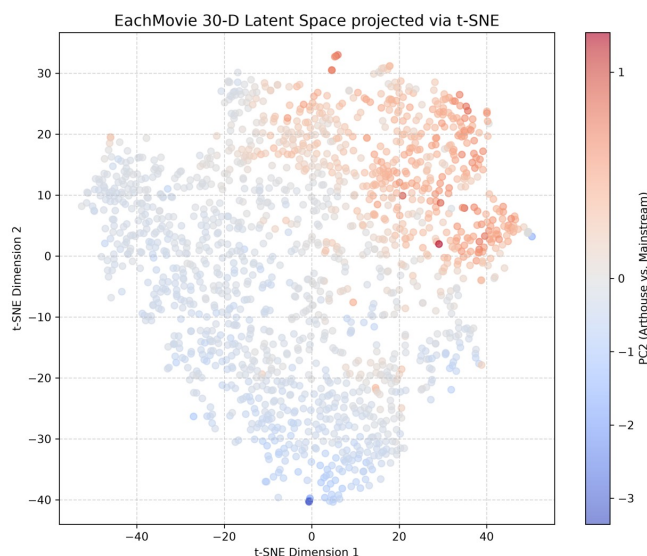
John DeTreville / DEC Systems Research Center (ret'd) / 2026-09-30 / Minneapolis, MN, USA

eachmovie.github.io

SLIDE 1

Hi, I'm John DeTreville. This is a historical talk about some work I did thirty years ago on a system called EachMovie. This is the “Past, Present, and Future” track and I picked “PAST.” BTW, NEVER call anything THE first, ESPECIALLY your own work! 

No t-SNE charts (not that kind of talk)



SLIDE 2

I don't have ANY tee-snee charts. It's not that kind of talk. (Maybe you're in the wrong room?)

But the paper does have a FEW numbers in it. READ THE PAPER if you want some numbers. [📄]

The 1990s

- Computers were about 1000× slower than now.
- The Web had only just begun.
- There were *almost no recommender systems*!
- And there were *no latent recommender systems —yet!*
- “We could change the world . . .”
 - . . . **NOT** !!! (At first, anyway. Read the paper.)

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SLIDE 3

Now, the Nineties were a MAGICAL
TIME [mime 🌈 two-handed] , even
if most of the folks HERE were hardly
even BORN yet! [👮]

Computers were slow. “The Wide, Wide World of the Web” was just “The Web.” There were VERY FEW

recommender systems, and there were
NO LATENT recommender systems. So I
figured, LET'S INVENT LATENT
RECOMMENDER SYSTEMS!

And so we did. We did it for movies. []

Guidance from Roger Ebert

(Very famous movie critic, *Chicago Sun-Times*)

- “The star rating system is relative, not absolute.
- “When you ask a friend if **Hellboy** is any good, you’re not asking if it’s any good compared to **Mystic River**, you’re asking if it’s any good compared to **The Punisher** . . .
- “If **Superman** is four, then **Hellboy** is three and **The Punisher** is two.”

Roger Ebert, “Shaolin Soccer”, 2004

SLIDE 4

With no other recommender systems to copy, I had to copy HUMAN recommender systems—like ROGER EBERT, very famous movie critic. So, how many ★s does a movie GET in his reviews? He said it should be RELATIVE, not absolute.

If he gave 3★ to, say, “Hellboy” —a comic-book movie—people would say, why did you give THAT 3★? HUUUH? But HE said, you want INDIVIDUALIZED RECOMMENDATIONS. And if SUPERMAN'S 4, then HELLBOY's 3, and THE PUNISHER is 2. (This is pretty darn close to a latent-space approach!) [

EachMovie (1995–97)

*“If you can name some movies you've liked . . .
we can name some others you might like too!”*

→US6078740A

- Grew to 1,628 movies, 72,916 users, 2,811,983 votes.
- The vote matrix was 97.63% sparse.
- Votes ranged from 0★ to 5★, plus Sounds Awful.

SLIDE 5

There's not much still SURVIVING from EachMovie, since it got SHUT DOWN in 1997 while the company was laser-focused on GOING OUT OF BUSINESS.

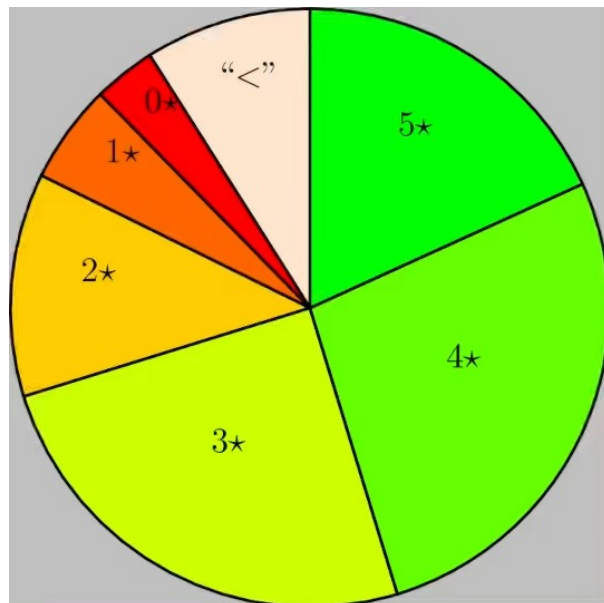
There's only a FEW artifacts extant, including an impressive-looking US

PATENT of mine, patenting [looks at patent] LATENT... RECOMMENDER... SYSTEMS... huh!—Applied 1996, Granted 2000, Expired 2016, so please feel free to use it now! (It overlapped the Netflix Prize, so half that money is rightfully mine!) [

EachMovie had about 2.8 million votes, and the vote matrix was **98% sparse**. The votes ran from 0★ to 5★, plus a special vote: “Sounds Awful,” which means, OMG, that “Sounds Awful!!!” [

Vote distribution and missing votes

- EachMovie's recorded votes were 9% **Sounds Awful** (" $<$ "), then 3% **0★**, 5% **1★**, 12% **2★**, 25% **3★**, 27% **4★**, and 18% **5★**.
- But 97.63% are missing from the vote matrix.
 - What to do, what to do?



SLIDE 6

What's "98% sparse" LOOK like? This pie chart SHOWS all those votes, but NOT SO MUCH, since most of the elements in the vote matrix were MISSING. Like THIS. [ pie chart recedes to 2.37% in the upper left, leaving a giant grey-scale question mark]

This FINAL FIGURE is of course to scale.
A LITTLE bit is in color (a little over 2%)
but ****98%**** of it is gray-scale. What to do,
what to do? []

Unfilled objective function

$$\sum_{(i,j)} (\text{Vote}_{ij} - \vec{u}_i \cdot \vec{v}_j)^2$$

K was 30

$$\sum_{(i,j) \in \Omega} (\text{Vote}_{ij} - \vec{u}_i \cdot \vec{v}_j)^2$$

$$\sum_{(i,j) \in \Omega} W_{ij} (\text{Vote}_{ij} - \vec{u}_i \cdot \vec{v}_j)^2$$

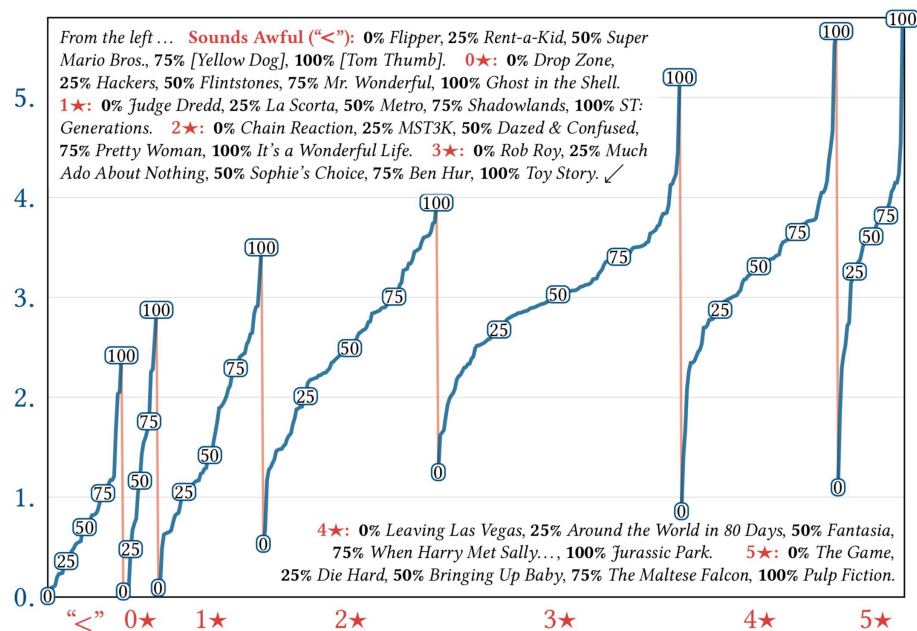
$$\sum_{(i,j) \in \Omega} W_{ij} (\text{Vote}_{ij} - \vec{u}_i \cdot \vec{v}_j)^2 + \lambda \left(\sum_i \|\vec{u}_i\|^2 + \sum_j \|\vec{v}_j\|^2 \right)$$

λ was 2.0 (or 0.08)

SLIDE 7

Here's some math that shows what you can do. • It's in the paper. • Read the paper. [

Giraffe chart



SLIDE 8

Let's look at MY 465 votes in EachMovie. Why MY votes? Because I had PERMISSION to use MY votes in these slides.

We bin them up by value: over there is "Sounds Awful," then 0★, 1★, 2★, up to 5★. We use my votes to draw the

GIRAFFE CHART so that we can... oh,
EFF IT, WRONG GIRAFFE CHART!!! [panic: original giraffe chart shows seven photorealistic giraffes with labels] [🔲 to reveal hidden chart]

Okay, OKAY, here's the RIGHT Giraffe Chart!!! (WHEW!) Look at the giraffe curve in the MIDDLE for 3★. I had about a HUNDRED votes I gave 3★. (This is all on GitHub: eachmovie.github.io.) We draw a nice little giraffe curve, and divide IT by QUARTILES. The MIDDLE is pretty flat but the OUTLIERS are “INTERESTING.” EachMovie says I SHOULD have ranked “Toy Story” HIGHER. Maybe it's right! [🔲]

User reactions

From: Anonymous

“Congratulations on your site, now I am using it to decide which movies to go to on weekends :)”

From: DB

“Amazing, I am going to have to get to the video store to check out some predictions.”

From: CC

“I love the service. I think that this is one of the coolest and most useful things I’ve ever found on the Internet. I give you my seal of approval.”

From: AK

“I enjoyed using your service and was pleasantly surprised to see some off-beat films listed -- *‘I Shot Andy Warhol’* and *‘Eat Drink Man Woman’* for example.”

From: RP

“This service is a brilliant idea and one that will indeed begin to define both the Web and marketing in the years to come.”

From: ??

“Just wanted to slip out a quick note- this is a greatly enjoyable site! It has lead us (my wife and myself) to some really great movie choices that we otherwise would have overlooked, but I guess that was the idea, wasn't it :)”

From: MDM

“I’ve been using EachMovie for about a month now and, I have to admit, you are pretty good in your predictions. Not all the time, of course you’re not God, but much of the time you’re close.”

SLIDE 9

There was ONE REALLY IMPORTANT THING about EachMovie: we got USER REACTIONS back from the users. It’s very hard NOWADAYS to find users of AUTOMATED RECOMMENDER SYSTEMS who’ve never SEEN an AUTOMATED RECOMMENDER SYSTEM before IN

• THEIR • ENTIRE • LIVES. But, you know, ****I**** found them! HA! Here's what some said. One person said: [🗨️]

YOU'RE NOT GOD, BUT MUCH OF THE TIME YOU'RE CLOSE. HA! Looks GREAT on my résumé! [🗨️]



Image: CC BY-SA 2.0 / Sound Opinions

Roger Ebert
(1942–2013)



© 1996 Digital Equipment Corporation

*"[EachMovie] proposes to suggest movies you might enjoy based on your opinions of movies you have seen. [...] To my absolute astonishment, the program evaluated my ratings and advised me I would probably enjoy **Pather Panchali**, **Before the Rain**, **Land and Freedom**, **Chungking Express**, and **Red**. This was *uncannily accurate* [...] The program was so much on the money, indeed, that I pictured a **Wizard of Oz**-type figure hiding behind the curtains, manually inserting suggestions, so I logged on again using a different address and password, and this time the program's recommendations included the splendid choices **Theremin**, **Fargo**, and **Crumb**. I am forced to concede that EachMovie works ... for me, anyway."*

Roger Ebert, Yahoo! Internet Life 2, 4 (Sep. 1996)

SLIDE 10



When Roger Ebert reviewed EachMovie, he said, WOW, it actually works! It's UNCANNILY • ACCURATE! Are they CHEATING? I'll launch a SYBIL ATTACK—yeah, STUPID COMPUTERS!— and “THEY”

won't be able to figure that out... right?

Well, EachMovie STILL worked! He compared EachMovie to _THE WIZARD OF OZ_, which ALSO looks great on my résumé, AND it means EachMovie PASSED THE TURING TEST! [ to expose bumper sticker]

So OF COURSE I made up BUMPER STICKERS—took his “UNCANNILY ACCURATE” & added an EXCLAMATION POINT to make it look even BETTER!!! I have some of these up here right now [ WAVE bumper stickers] that you can come up and ASK for, if you have some MONEY. [

EachMovie's Top Ten

10. *Aladdin* (228K★)
9. *The Shawshank Redemption* (231K★)
8. *Forrest Gump* (235K★)
7. *The Silence of the Lambs* (242K★)
6. *True Lies* (263K★)
5. *Batman* (263K★)
4. *The Fugitive* (266K★)
3. *Pulp Fiction* (304K★)
2. *Dances With Wolves* (320K★)
1. *Apollo 13* (329K★)



All third-party image sources for these crudely drawn derived works are in the public domain or otherwise freely available in the United States. For further details, see [Coyote vs. ACM \(2026\)](#).

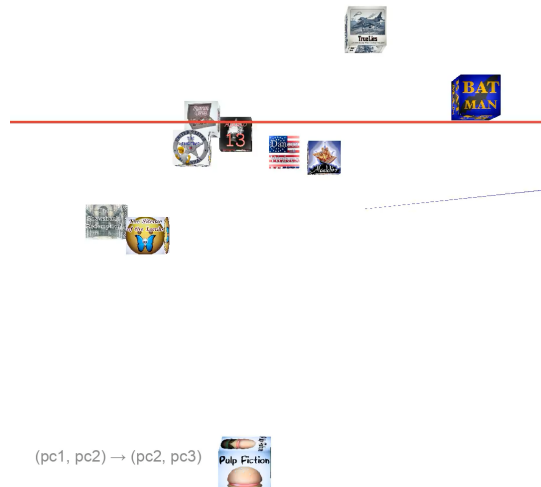
SLIDE 11

Let's get ready for the fabulous EachMovie MUSIC VIDEO! Here's the TOP TEN movies in EachMovie, based on how many K★ they got total. We got *_Aladdin_*, *_The Shawshank Redemption_*, all the way up to *_Apollo 13_*. These are all VERY different movies, so let's PLOT them—

not the EASY way but the HARD way!
BTW, these icons are VERY CRUDELY
DRAWN.

BTW, look at the FOOTNOTE! You should
ALL go see THIS year's FEEL-GOOD
COMEDY HIT OF THE SUMMER™,
Coyote vs. ACM (2026). [

PC 1&2&3→2&3&4 projected through 30-space



SLIDE 12

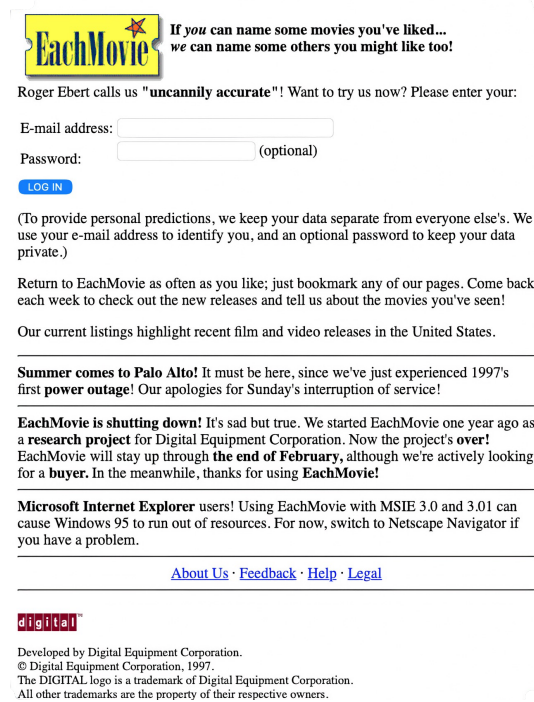
Here, we take Principal Components 1, 2, and 3. We PROJECT all those icons onto a rotating 3-D subspace. (It's a PERSPECTIVE plot.) When you do the math RIGHT, here's what you get. [📺]

OOH, AAH. And I have a BUNCH of

this video; we should watch it all together
later on. It's FASCINATING. [❤️] **|**
LOVE it! [📺]

EachMovie (old)

- EachMovie shut down in late 1997.
- The EachMovie dataset was available to researchers until 2004 and was widely used.
- Top social-media post about EachMovie: *“Where can I find the original EachMovie dataset? (for a research paper I’m working on)”*



The screenshot shows the EachMovie website's login page. At the top, there is a yellow banner with the EachMovie logo and a quote: "If you can name some movies you've liked... we can name some others you might like too!". Below this, a message states: "Roger Ebert calls us 'uncannily accurate'! Want to try us now? Please enter your:". There are two input fields: "E-mail address:" and "Password:" (with "(optional)" next to it). A blue "LOG IN" button is positioned below the password field. A note in parentheses explains: "(To provide personal predictions, we keep your data separate from everyone else's. We use your e-mail address to identify you, and an optional password to keep your data private.)". Below the login fields, there are several paragraphs of text: "Return to EachMovie as often as you like; just bookmark any of our pages. Come back each week to check out the new releases and tell us about the movies you've seen!", "Our current listings highlight recent film and video releases in the United States.", "Summer comes to Palo Alto! It must be here, since we've just experienced 1997's first power outage! Our apologies for Sunday's interruption of service!", "EachMovie is shutting down! It's sad but true. We started EachMovie one year ago as a research project for Digital Equipment Corporation. Now the project's over! EachMovie will stay up through the end of February, although we're actively looking for a buyer. In the meanwhile, thanks for using EachMovie!", and "Microsoft Internet Explorer users! Using EachMovie with MSIE 3.0 and 3.01 can cause Windows 95 to run out of resources. For now, switch to Netscape Navigator if you have a problem." At the bottom, there is a navigation bar with links: "About Us · Feedback · Help · Legal". Below the navigation bar is the "digital" logo and a footer containing the text: "Developed by Digital Equipment Corporation. © Digital Equipment Corporation, 1997. The DIGITAL logo is a trademark of Digital Equipment Corporation. All other trademarks are the property of their respective owners."

SLIDE 13

EachMovie lasted through 1997, and then got SHUT DOWN while DEC went out of business, funny story.

The EachMovie dataset was available until 2004, another funny story.

Look at the slides online to see some

cool details! But NOW, EachMovie's
GONE. [ to erase screenshot]

Bye bye. [

EachMovie (new)

- A resurrected EachMovie website is planned for researchers and historians.
- Check the EachMovie blog at eachmovie.github.io for updates.
- *Careful: the website may be running on an old Mac Mini in my garage.*



SLIDE 14

But I have the SOURCE CODE, and the ORIGINAL DATASET (because someone FOUND them in their garage, and MAILED them to me. *THANKS*!) So I RECONSTRUCTED it, and HERE'S what you get! [📺 to show new screenshot]

Those latent factors computed back in the 1990s STILL WORK. **MATH DOESN'T DIE.** That's the NEAT THING. And the code even prints out COPYRIGHT 1996, DIGITAL EQUIPMENT CORPORATION. [



Epigraph / epilogue

I have heard the mermaids singing, each to each.

I do not think that they will sing to me.

— “The love song of J. Alfred Prufrock,” T.S. Eliot, 1917

- So, I figured, let's **do** something about that!
And that's the story of EachMovie.

SLIDE 15

Here's a closing EPIGRAPH, also EPILOG, so you get two for one! T.S. ELIOT, Very Famous Twentieth Century Modernist Poet, wrote: [reads] “I HAVE HEARD THE MERMAIDS SINGING, EACH TO EACH” —that's where the name COMES from, EachMovie. [repeats

] _“Mermaids singing each to each,”_ [reads] “I • DO • NOT • THINK • THAT • THEY • WILL • SING • TO • ME.” [😊]
] Well, that seems SAD, doesn't it? [👤]
]

And so I THOUGHT, let's **DO** something about that! And THAT'S THE STORY OF EACHMOVIE, thankyouverymuch. [👤]

Past, Present, and FUTURE

- **PAST:** EachMovie factored the vote matrix to expose the latent reality underneath. Users **wanted** to use EachMovie. They sought it out. They gladly shared their tastes & souls. EachMovie returned the favor by **letting them drive**.
- **PRESENT:** Those carefree halcyon days of the 1990s are long gone. Now our modern dystopia dials the **doomscrollamine** ^[1] all the way up to 11.
- **FUTURE:** “There are two paths we can go by.” ^[2] And there’s still time to change the road we’re on.

SLIDE 16

EDITORIAL. (The views expressed here are not those of Digital Equipment Corporation.) [much more deliberately] EachMovie was the FIRST • LATENT • RECOMMENDER • SYSTEM. It gave its human USERS human AGENCY. They WANTED to use EachMovie, and

to refine its MODELS of their tastes.
EachMovie LET *THEM* DRIVE.

As I said, we let users WRITE IN, and many DID. One user wrote that SHE ran EXPERIMENTS with HER EachMovie account, TWIDDLING the KNOBS. She'd change one or two of her votes then click **Submit** and SEE WHAT HAPPENED. (EachMovie updated predictions instantly.) Then she'd change them BACK, watch her predictions snap back into place, THEN try something NEW. This is a TEXTBOOK EXAMPLE of HUMAN AGENCY. We didn't plan for this—it's an EMERGENT PROPERTY—but this KIND of HUMAN AGENCY would SEEM like something you'd want to TRY to REPLICATE in modern systems.

But THAT • was • THEN • and • THIS •
is • NOW. **NOW** we're on a
DIFFERENT timeline. We're STUCK in
BACK TO THE FUTURE PART II (1989)
and BIFF TANNEN is RUNNING THE
CASINO! *HE'S* **DIALING** the
DOOMSCROLLAMINE all the way **UP
TO 11!** (BIFF is the BAD GUY in the
movie!)

So, what to do, what to do? [move
hands to mime a balance] THESE are
the TWO PATHS WE CAN GO BY. And, IN
THE LON— NO, **NO**, no, RIGHT
NOW! —IT'S • NOT • TOO • LATE •
TO • CHANGE • THE • ROAD • WE'RE
• ON.

Makes you wonder. Yeah. Yeah. Really,
really makes you wonder. [🗨️]

Extended Summary

- a) EachMovie let users drive.
- b) Doomscrollamine is bad. [1]
- c) **There's still time to change the road you're on.** [2]

QUESTIONS?

[1] Andrew Marltton, 2026.

[2] Robert Plant, 1971.

EachMovie's Top Ten Movies

"Across 30 Dimensions"

—A Music Video—

by John DeTreville, 2026

SLIDE 17

[Stand Up Straight and face
audience, arms akimbo, fists on waist
] [😊😊😊]

So NOW it's time for QUESTIONS!
ANYONE? This fabulous SILENT
MUSIC VIDEO will help EVERYONE
think of EASY QUESTIONS with EASY

ANSWERS!!!

[] [move downstage center as video starts. as music plays, strip off gray embroidered overshirt to expose bright orange T-shirt emblazoned with GOT QUESTIONS?, then return to lectern just as brief music fades while video plays on]

[] Here are some TOPICS for CONVERSATION! Anyone?

[take questions]

["be sure to LIKE and SUBSCRIBE !!!"]

[exit _as if_ "pursued by a bear"
(Winter's Tale, Act 3, Scene 3); the bear

will not be taking questions]